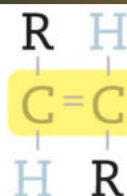




UNIT 03

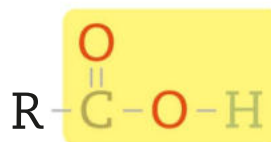
ORGANIC CHEMISTRY



Alkenes have the functional group $\text{C}=\text{C}$.



Alcohols have the functional group $-\text{OH}$.



Carboxylic acids have the functional group $-\text{COOH}$.



Esters have the functional group $-\text{COO}-$.

ORGANIC COMPOUND

Organic compounds are carbon-based chemicals found in living organisms and non-living things. They are important for life processes and used in various everyday products.

HISTORY OF ORGANIC CHEMISTRY

EARLY BELIEFS

Before the 18th century, people believed organic compounds could only be produced by living organisms (Vital Force Theory by Berzelius). Stated that:

"Organic compounds can only be synthesized by living organisms as they possess a special 'vital force' that cannot be recreated in a laboratory."

This theory was later disproved by Friedrich Wöhler in 1828 when he synthesized urea (an organic compound) from ammonium cyanate (an inorganic compound).

REPRESENTATION OF ORGANIC COMPOUND

Organic compounds have four different sorts of formulae:

- Molecular formulae
- Structural formula
- Condensed formula
- Dot and Cross formula

Previous Year's Questions



State vital force theory and how was this theory discarded by Friedrich Wohler?

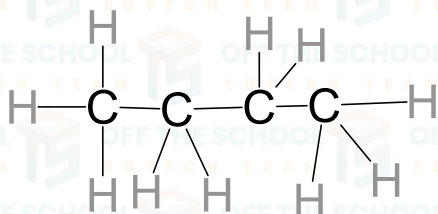
[2023-24]

I. MOLECULAR FORMULA

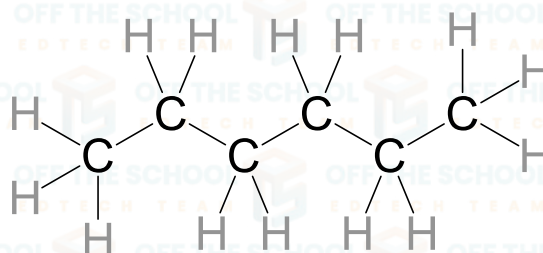
A chemical formula that represents the number and types of atoms present in a molecule. For Example C_4H_{10} (Butane) and C_6H_{12} (Hexane).

II. STRUCTURAL FORMULA

A diagram that shows the arrangement of atoms in a molecule and the bonds between them.



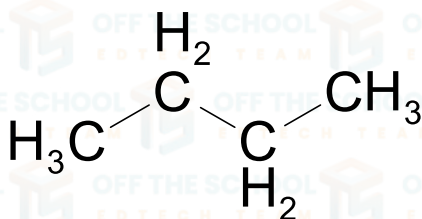
Butane



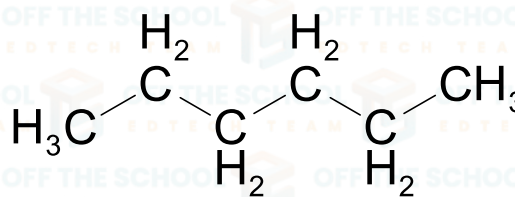
Hexane

III. CONDENSED FORMULA

A shorter version of the structural formula that omits some bond lines and simplifies the representation of a molecule.



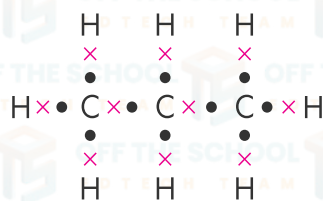
Butane



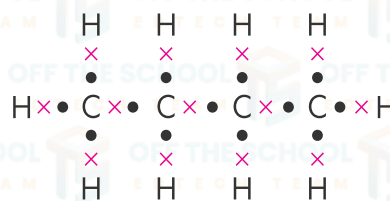
Hexane

IV. DOT AND CROSS FORMULA

A representation of a molecule that uses dots to represent valence electrons and crosses to represent positive charges or lone pairs of electrons.



Propane



Butane

General Properties of Organic Compound

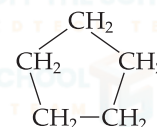
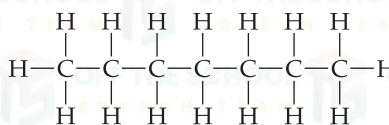
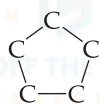
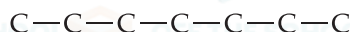
Property	Organic Compounds
Bonding	Covalent
State at room temperature	Liquid, gas, or solid
Boiling points	Low
Melting points	Low
Solubility in water	Most are not soluble in water
Solubility in inorganic solvents	Soluble
Conduct electricity in aqueous solution	No
Combustion	Most burn
Reaction rates	Slow

DIVERSITY & MAGNITUDE ORGANIC COMPOUND

The following factors contribute to the existence of such a great number of organic compounds:

1. CATENATION

It is the ability of carbon atoms to form long chains and bond with each other, resulting in the vast diversity of organic compounds.



2. ISOMERISM

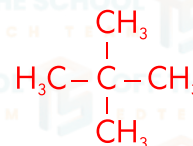
It refers to the phenomenon where different organic compounds have the same molecular formula but differ in the arrangement or connectivity of atoms, leading to distinct chemical and physical properties.



n-pentane



iso-pentane



neo-pentane

3. CARBON'S COVALENT BOND STRENGTH

Carbon forms strong covalent bonds with other elements, allowing it to create stable and diverse organic compounds that play a crucial role in the chemistry of life.

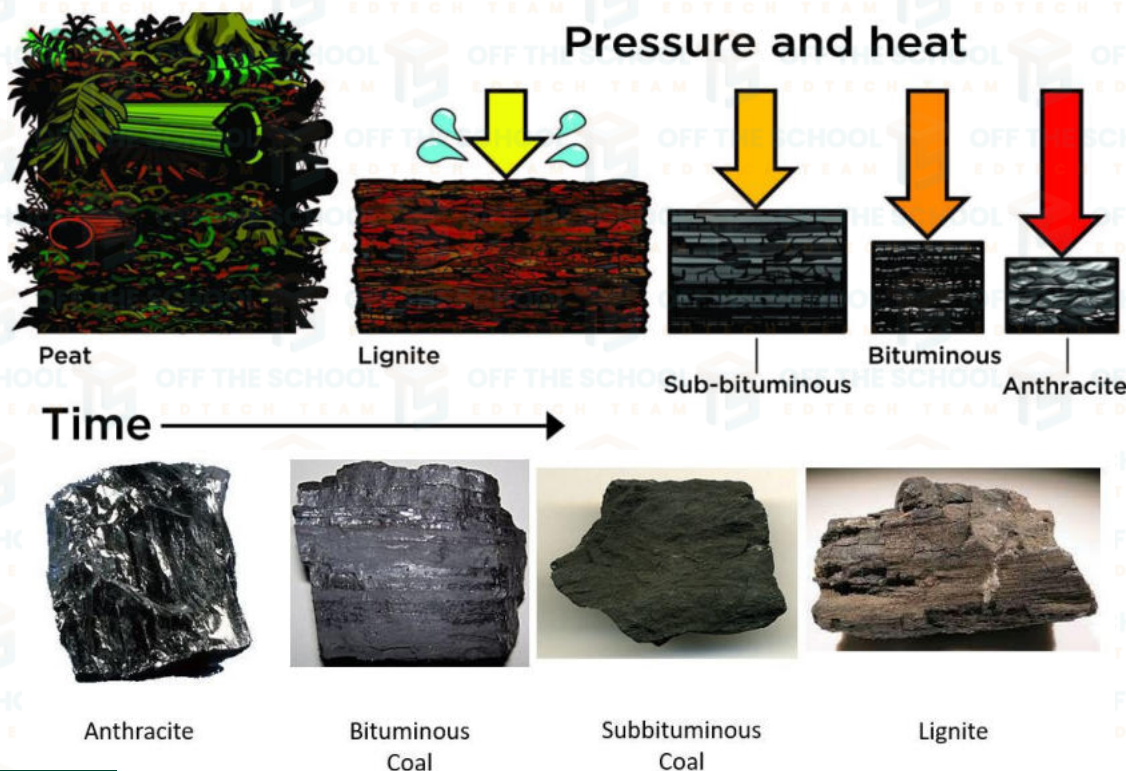
4. MULTIPLE BONDING

Carbon can form multiple bonds, such as double or triple bonds, with other atoms, enabling the formation of complex organic molecules with unique reactivity and properties.

SOURCES OF ORGANIC COMPOUND

1. COAL

Organic compounds can be found in coal, which is a fossil fuel formed from the remains of ancient plants and organic matter that underwent geological processes over millions of years.



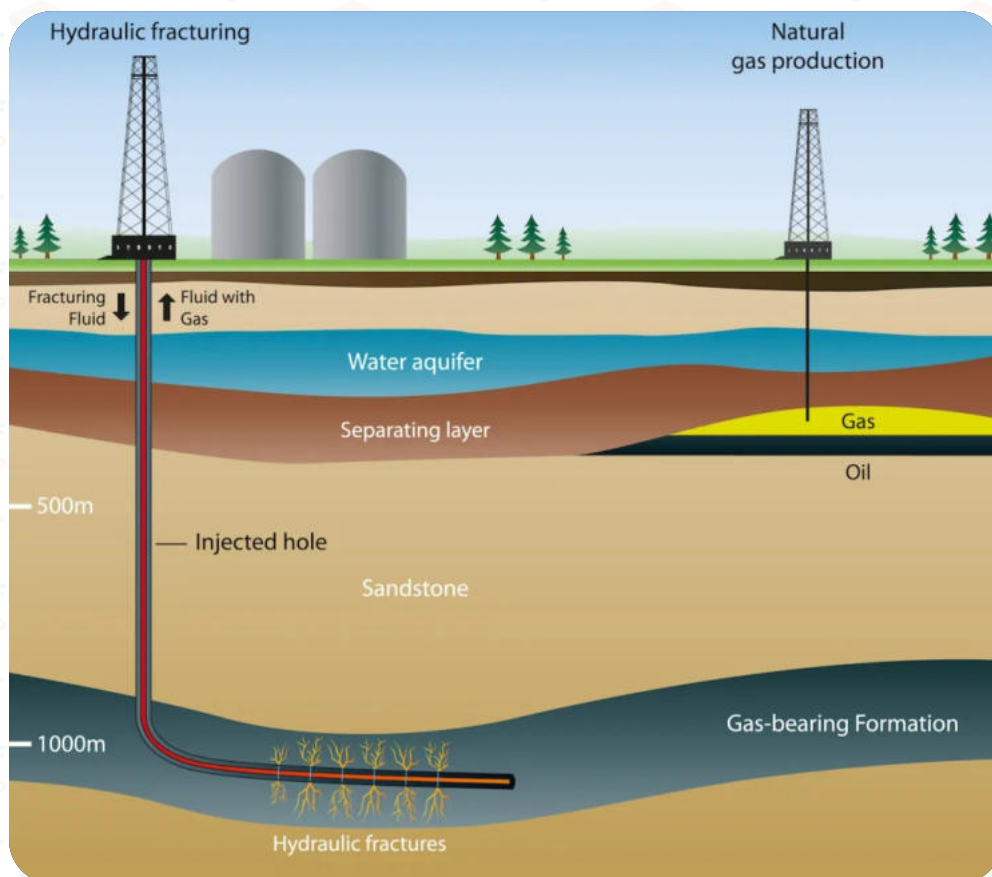
2. PETROLEUM

Organic compounds are abundant in petroleum, a naturally occurring mixture of hydrocarbons that is formed from the decomposition of marine organisms over millions of years and extracted from underground reservoirs.



3. NATURAL GAS

Organic compounds are present in natural gas, a fossil fuel composed mainly of methane (and other hydrocarbons), formed from the decomposition of organic matter in ancient deposits of plants and microorganisms.



4. PLANTS

Organic compounds are synthesized by plants through photosynthesis, where they convert carbon dioxide and water into glucose and other organic molecules, making plants a significant source of organic compounds in nature.

5. SYNTHESIS IN LABORATORY

Organic compounds can also be synthesized in the laboratory through various chemical reactions, allowing researchers to create specific organic compounds for a wide range of applications in industries such as pharmaceuticals, materials, and agriculture.

SOURCES OF ORGANIC COMPOUND

A homologous series is a group of organic compounds in which each adjacent member is differ by a CH_2 unit in their molecular formula.

The First 10 Continuous-Chain Alkanes

Name	Molecular Formula	Condensed Structural Formula
Methane	CH_4	CH_4
Ethane	C_2H_6	CH_3CH_3
Propane	C_3H_8	$\text{CH}_3\text{CH}_2\text{CH}_3$
Butane	C_4H_{10}	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
Pentane	C_5H_{12}	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
Hexane	C_6H_{14}	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
Heptane	C_7H_{16}	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
Octane	C_8H_{18}	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
Nonane	C_9H_{20}	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
Decane	$\text{C}_{10}\text{H}_{22}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

SATURATED HYDROCARBONS	UNSATURATED HYDROCARBONS
1. Contains only single bonds between carbon atoms.	Contain at least one double or triple bond between carbon atoms.
2. Have the maximum possible number of hydrogen atoms.	Have fewer hydrogen atoms due to double or triple bonds.
3. Less reactive compared to unsaturated hydrocarbons.	More reactive due to the presence of multiple bonds.
4. Generally, have higher boiling points.	Tend to have lower boiling points than saturated hydrocarbons.
5. Examples include alkanes (e.g., methane, ethane).	Examples include alkenes (e.g., ethene, propene) and alkynes (e.g., ethyne, propyne).
6. Are relatively stable and less likely to undergo chemical reactions.	Can undergo addition reactions due to the presence of multiple bonds.
7. C_nH_{2n+2} is the general formula.	C_nH_{2n} is the general formula for alkenes, and C_nH_{2n-2} for alkynes.

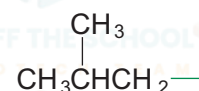
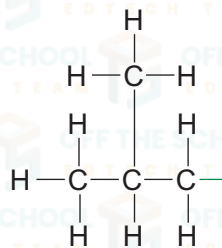
ALKYL RADICAL

When one hydrogen atom from the alkane is removed then it is known as Alkyl radical. It is represented by "R". The general formula is C_nH_{2n+1}

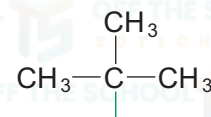
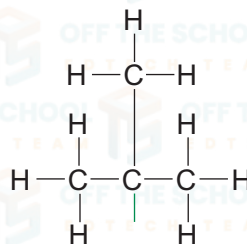
Common Alkyl Groups

	Name	Structural Formula	Condensed Structural Formula
Derived from Propane	Propyl	$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}- \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$	$\text{CH}_3\text{CH}_2\text{CH}_2-$
	Isopropyl	$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & \\ \text{H} & & \text{H} \end{array}$	CH_3CHCH_3
Derived from Butane	Butyl	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}- \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-$
	Secondary butyl (sec-butyl)	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \\ \text{H} & & \text{H} & \text{H} \end{array}$	$\text{CH}_3\text{CHCH}_2\text{CH}_3$

Derived from Isobutane Isobutyl



Tertiary butyl
(*tert*-butyl)



FUNCTIONAL GROUPS

The active part of an organic compound is actually a Functional group.

FUNCTIONAL GROUPS CONTAINING CARBON, HYDROGEN, AND OXYGEN

1. ALCOHOLIC GROUP

It consists of a hydroxyl group ($-\text{OH}$) attached to a carbon chain, found in alcohols like ethanol ($\text{C}_2\text{H}_5\text{OH}$).

2. ETHER LINKAGE

It involves an oxygen atom bonded to two carbon groups, present in ethers such as dimethyl ether ($\text{CH}_3-\text{O}-\text{CH}_3$).

3. ALDEHYDIC GROUP

It contains a carbonyl group ($\text{C}=\text{O}$) at the end of a carbon chain, as seen in aldehydes like formaldehyde (HCHO).

4. KETONIC GROUP

It features a carbonyl group ($\text{C}=\text{O}$) within the carbon chain, found in ketones like acetone ($\text{CH}_3-\text{CO}-\text{CH}_3$).

5. CARBOXYL GROUP

It consists of a carbonyl group ($\text{C}=\text{O}$) and a hydroxyl group ($-\text{OH}$) on the same carbon, present in carboxylic acids like acetic acid (CH_3COOH).

6. ESTER LINKAGE

It involves a carbonyl group ($\text{C}=\text{O}$) bonded to an oxygen atom, which is in turn bonded to a carbon chain, present in esters like ethyl acetate ($\text{CH}_3-\text{COO}-\text{C}_2\text{H}_5$).

FUNCTIONAL GROUPS CONTAINING CARBON, HYDROGEN, AND NITROGEN

AMINES

Amines are organic compounds derived from ammonia (NH_3) by replacing one or more hydrogen atoms with alkyl ($-\text{R}$) or aryl ($-\text{Ar}$) groups.

GENERAL FORMULA:

Primary (1°) Amine: $\text{R}-\text{NH}_2$

Secondary (2°) Amine: R_2-NH

Tertiary (3°) Amine: R_3-N

EXAMPLE:

Methylamine (CH_3NH_2) $\longrightarrow$ Primary Amine
 Dimethylamine (CH_3NHCH_3) $\longrightarrow$ Secondary Amine
 Trimethylamine ($\text{N}(\text{CH}_3)_3$) $\longrightarrow$ Tertiary Amine

Important Functional Groups in Organic Compounds

Functional Group	Compound Type	Prefix or Suffix of Name	Lewis Structure	Example	Systematic Name (Common Name)
	alkene	-ene			ethene (ethylene)
	alkyne	-yne			ethyne (acetylene)
	alcohol	-ol			methanol (methyl alcohol)
	haloalkane (X = halogen)	halo-			chloromethane (methyl chloride)
	amine	-amine			ethanamine (ethylamine)
	aldehyde	-al			ethanal (acetaldehyde)
	ketone	-one			2-propanone (acetone)
	carboxylic acid	-oic acid			ethanoic acid (acetic acid)
	ester	-oate			methylethanoate (methyl acetate)
	amide	-amide			ethanamide (acetamide)
	nitrile	-nitrile			ethanenitrile (acetonitrile, methyl cyanide)